

HW 1.2b

Name(s): _____

1. Imagine that you take the cube of a certain number. To restore its original value, _____ it.

2. Solve the following equation, by combining the two exponents.

$$N = 10^{-15} \times 10^{-20}$$

3. Neptune's year is 164.8 times that of Earth. How long is this in seconds?

4. Simplify: $(10^a)^3$

5. What is the value of R in the equation shown below, if $L = 576$ and $T = 2$?

$$R = \frac{\sqrt{L}}{T^2}$$

6. A Tardigrade (a microscopic creature) has a length of about $400 \mu\text{m}$.

A. How big is this in mm? in cm?

- B.** If you put tardigrades in a line, one after another, how many would fit in one meter?
- 7.** How old are you in years, days, and seconds?
- 8.** Simplify $10^a \times 10^b$
- 9.** Solve the equation $x^4 = 10^3$
- 10.** How many seconds are in one year?
- 11.** One Astronomical Unit (AU)=149,597,870,700 meters. One Light Year (ly) = 9,460,730,472,580,800 meters.
- A.** Write 1 AU in Scientific notation, rounding after the third number.
- B.** Write 1 ly in Scientific notation, rounding after the third number.
- C.** How many times longer is 1 ly than 1 AU?

- 12.** The equation that calculates the drag force (F_d) on a suspended spherical particle is

$$F_d = 6\pi v^{1.5} r^2$$

Rearrange the equation to solve for v in terms of the other variables.

- 13.** Solve for x : $x^{3.2} = y^{6.7}$

- 14.** Simplify: $\frac{10^a}{10^b}$

- 15.** If the Earth has a radius of $r_{\oplus} = 6.37 \times 10^6$ meters.

A. What is the circumference of the Earth in meters? kilometers?

B. What is the Volume of the earth in meters³

16. How many times longer is one *Gigayear* (Gyr) than one *nanosecond* (ns)?

17. What is the value of L in the equation shown below, if $R = 0.2$ and $T = 3$? $L = R^2T^4$

18. Solve the following equation, by evaluating the fourth root. $\boxed{\sqrt[4]{2401}}$

19. Simplify the following:

A. $10^2 \times 10^3$

B. $10^2 \div 10^3$

C. $(10^2)^3$

20. How many minutes are in one day? How many minutes are in one year?